

What are the most important factors in determining the level of volatility a geopolitical event causes in global equity markets?

Abstract

This project seeks to investigate the single most important factor in determining the level of volatility a geopolitical event causes in global equity markets. Throughout the dissertation, I aim to weigh up and rank market liquidity, predictability and commodity dependency by importance in influencing equity market volatility. I found that each factor determines a distinct stage of the volatility process with commodity dependency largely influencing the initial magnitude of volatility where the event threatens essential factor inputs. Predictability determines whether information is incorporated gradually or through quick repricing and liquidity determines whether the resulting trading pressure is absorbed or amplified into a persistent volatility regime.

I evaluate this ranking using economic and financial-market theory combined with four major case studies and original event-study regressions. Using the Russia-Ukraine conflict, I unpick how reliance on energy exports can magnify equity market consequences, notably affecting Europe much more than the US and Japan. By contrast, the Iraq War demonstrates that the significance of commodity dependence diminishes over time as investors assume long-run supply stability, thus, limiting persistent volatility. The use of hedging mechanisms and futures further mitigates persistent volatility through reduced panic selling.

I then compare Trump-era tariffs with Brexit in order to highlight the role of predictability and liquidity. Under Trump's erratic policymaking, volatility persisted longer because his indecisiveness produced repeated structural breaks in volatility. However, Brexit's somewhat sequenced and partly anticipated series of events generated a shorter-lived equity impact, although still meaningful. Carrying out a cross-sectional regression of S&P 500 stocks during April 2025, I found that liquidity variables explained 57.2% of cross-sectional volatility versus 7.3% for policy-uncertainty proxies, therefore, supporting the central claim that liquidity acts as the primary transmission channel.

Liquidity emerges as both an independent factor and the channel through which predictability and commodity dependence affect prices. Additionally, less developed markets are more vulnerable to lower liquidity levels because of their weaker central-bank buffers. On the contrary, developed markets are more cushioned because of their well-established and well-equipped central banks, yet this cushion is itself part of how liquidity governs volatility. Therefore, I find that the most convincing conclusion is that no single factor explains every event given the complex nature of equity markets, but liquidity arises as the most important transmission mechanism, predictability is the

deepest determinant of persistence, and commodity dependence is a conditional amplifier of initial magnitude that is incorporated in specific geopolitical events.

1. Introduction

As globalisation continues to evolve in an increasingly integrated global economy, the influence of geopolitical events on global equity markets has rarely been more prominent. Most recently as in April 2025 when President Trump's tariff announcements shocked every major equity index within hours. Therefore, understanding precisely what it is that differentiates geopolitical events from generating prolonged periods of volatility compared to rapidly diminishing volatility regimes is crucial when attempting to analyse how investors manage risk across global portfolios and how policymakers dictate market conditions during periods of stress.

Intuitively, the economic weight of the countries involved, or the suddenness of an event might appear to be the most important factors. However, such broad categories offer little explanatory power compared with more nuanced indicators that not only correlate with volatility but actively drive it. This dissertation focuses on three such indicators: commodity dependence, event predictability, and market liquidity.

1.1 What "most important" means in this dissertation

Given this question focuses on achieving a ranking of several factors, the criteria for "importance" must be made clear before I analyse such factors. Across this dissertation, I investigate each factor against four tests: (i) initial magnitude, how large the first volatility move is; (ii) persistence, how long volatility persist for after the shock; (iii) transmission power, whether the factor amplifies or dampens other factors; and (iv) consistency across cases, whether the factor has a significant impact across different types of event.

Across these four tests, I argue that liquidity is the most important overarching factor given it governs the transition from the initial shock to the prolonged period of volatility, acting as the channel through which the other two factors reach prices. Predictability is the second most important factor because of how it shapes the speed and disorder of full information incorporation into markets, but its effect is strongest when it operates through disruptive liquidity conditions. Commodity dependence is highly important for the initial magnitude of volatility; however, it is less reliable as an explanation of persistence because markets are able to overcome such issues through hedging, substituting supply, or pricing in long-run supply stability given future expectations. This is the ranking I go on to defend across the rest of the dissertation.

1.2 Method in brief

My argument is built from three forms of evidence used in combination with each other. I incorporate established economic and financial theory, four event-study case studies (Russia-Ukraine 2022, Iraq 2003, Trump tariffs 2024–25, Brexit 2015–20), and original regression analysis using publicly available market data. The empirical work carried out is used to investigate whether the theoretical mechanisms investigated across academic papers holds in market data and translates into visible results. The purpose of my empirical findings is not to claim that any one variable mechanically determines volatility in isolation and the reasons for that caution are spelled out in Section 2.

2. Methodology and source evaluation

2.1 Source evaluation

For this project, I drew from the three categories of evidence, treating each distinct one differently as they each served differing purposes. I used peer-reviewed academic literature primarily to deepen my understanding of economic theory and mechanisms. This is because it is methodologically explicit and has been formally reviewed by credible academics. However, many of the papers I draw on are not written as to address my exact research question. Instead, I use them as theoretical building blocks rather than as direct proof. I used market and macroeconomic datasets such as: CBOE, Eurex, Yahoo Finance, LSEG, IEA, IMF, World Bank and Federal Reserve, to measure volatility, exposure and liquidity, but the conclusions drawn from them depend heavily on the proxy chosen and the event window used. I therefore state both explicitly throughout and ensure equity market data is cited from credible financial sources that are widely used by professionals. Thirdly, I utilise financial journalism, principally the Financial Times, for event chronology and contemporaneous market interpretation. Importantly not as proof of causality, because of the opinionated nature of articles that mean they are often not peer-reviewed and may reflect the framing of the publication rather than factual evidence. Where possible I cross-check factual claims across more than one source type before using them in the argument. A full reliability table is given in Appendix B.

2.1.2 Practitioner conversations

In addition to written sources, I conducted two informal expert conversations with finance-sector professionals to ensure my interpretations of the importance of liquidity translated into real institutional settings. The first was with a former Neuberger Berman employee whose role had involved working with official institutions including central banks, giving me valuable insight into their everyday operations and macroeconomic views. The second was with a finance-sector professional with broader markets

experience as a trader. The full record (date, role, themes, limitations, how the evidence informed the dissertation) is in Appendix A. I treat these conversations as qualitative expert testimony rather than statistical evidence: they are subjective, non-replicable, and could be biased by each individuals' professional backgrounds. As a result, I do not refer to them as concrete evidence but supporting/evaluative views that improve the quality of my investigation. They are used in the liquidity section as supporting practitioner context for central bank backstops, not as proof of causality.

2.2 Data sources and variables –

Citations are made under relevant figures and regressions referencing each mentioned source.

Equity-market data was drawn from Yahoo Finance, investing.com and Macrotrends for daily index closes on the S&P 500, Euro Stoxx 50, DAX, Nikkei 225 and FTSE 100.

Volatility was measured using three indicators: the VIX (CBOE), the VSTOXX (Eurex), and 21-day rolling realised volatility computed from index returns. VIX and VSTOXX are implied-volatility indices derived from option prices; they measure expected, not realised, volatility, and they also reflect global risk appetite, so I report them alongside realised volatility rather than relying on either alone.

Commodity-dependence data, particularly Russian gas dependency by country, was drawn from the IEA, World Bank and IMF World Economic Outlook. National energy dependence is an imperfect proxy for the exposure of a listed equity index. This is because index composition can over- or under-represent energy-sensitive sectors. I make sure to flag this proxy mismatch wherever the figures are used.

Predictability was proxied by the Economic Policy Uncertainty (EPU) index, forecast dispersion in oil futures, and news-count indices (Louis, 2026b). The oil-futures dispersion figures originate from Leigh, Wolfers and Zitzewitz paper (Leigh, 2003).

Liquidity was measured using bid-ask spreads, the Amihud illiquidity ratio, and trading volume, drawn from LSEG and WRDS where available (Amihud, 2020).

2.3 Variables and empirical method

The core empirical approach I took to investigate volatility is an event-study comparison of volatility during a defined post-event window with a pre-event baseline volatility level, supported by cross-sectional and time-series regressions. The dependent variable is the change in volatility relative to baseline ($\Delta\sigma$), supplemented by ICSS structural-break counts and GARCH conditional variance estimates.

A complete cross-sectional model would take the form:

$$\Delta Vol_i = \alpha + \beta_1 \cdot Dependence_i + \beta_2 \cdot Predictability_i + \beta_3 \cdot Liquidity_i + \Gamma' \cdot Controls_i + \varepsilon_i$$

In practice, I did not include all explanatory variables in every regression. Instead, I tested one or two at a time within each case study, then estimated the full multivariate model once for evaluation in Section 9. Time-series tests (GARCH and the Inclán–Tiao ICSS algorithm, implemented via the R ICSS package (Köstlmeier, 2021), are used to assess persistence and to detect structural breaks in variance; for which code is provided in Appendix A

To sufficiently answer my question, I used a combination of theoretical analysis, case study findings and limited regression analysis so that the accuracy of my conclusion would not be limited by the biases of any one method. Given the complex and varying nature of geopolitical events, I concluded that any single statistical approach would likely over simplify the relationship between volatility and the factors I test, which risks giving me a false level of precision. My regression results are therefore used as supporting evidence rather than as definitive proof of causality. This matters because the variables are not entirely independent. For instance, lower predictability can often reduce liquidity, while commodity exposure can increase uncertainty about earnings. The aim of my empirical work was predominantly to test whether the theoretical mechanisms are visible in market data, rather than to claim that one variable mechanically is solely responsible for volatility in isolation.

3. Key definitions

Term	Definition
Equity-market volatility	Volatility is the rate at which the price of a stock increases or decreases over a particular period. (Fidelity, 2026)
Realised volatility	A model-free measure of actual return variation, computed as the square root of the sum of squared intraday returns sampled at high frequency over a fixed window (Andersen, 2003).
VIX / VSTOXX	A volatility index like the VSTOXX® measures the expectancy of future stock index movements. Time horizons for standard deviation (also known as volatility) scale with the square root of time. So, to translate a VSTOXX® value of 20 into how much the market expects the SX5E to move in the next 30 days (about 1/12th of a year), we divide by the square root of 12, or around 3.5. This means that when the VSTOXX® is at 20, the market expects: • A roughly 65% chance that the index will be within a ±5.7% range one month from now

Term	Definition
	A roughly 95% chance that the index will be within a $\pm 11.4\%$ range one month from now (Management, 2021).
Commodity dependence	The condition in which a country's export revenues, growth, or equity market returns are predominantly driven by primary commodity prices, making the economy highly exposed to external price shocks (Nations, 2023).
Event predictability	The extent to which investors can anticipate the timing, probability, and likely economic consequences of an event before it materialises, determining how much is already priced in at the announcement date (Caldara, 2022).
Liquidity	A multidimensional property of a market reflecting how quickly and cheaply assets can be traded without generating a significant price impact. Kyle (1985) identifies three dimensions: tightness (cost), depth (volume), and resilience (speed of recovery) (Kyle, 1985).
Bid–ask spread	The difference between the lowest price at which a seller will accept and the highest price a buyer will pay. It reflects transaction costs and compensates dealers for order-processing, inventory, and adverse-selection risk (Glosten, 1985).
Amihud illiquidity ratio	A price-impact measure defined as the daily ratio of a stock's absolute return to its dollar trading volume, averaged over a period. It approximates how much price movement is associated with one dollar of trading activity (Amihud, 2020).
Inventory Risk	The risk incurred by a market maker who holds a non-zero inventory position in a security, leaving them exposed to adverse price movements until the position is unwound (Ho, 1981).
Market depth	The volume of buy and sell orders available near the prevailing market price. A deep market can absorb large orders with minimal price impact; a shallow market cannot (Kyle, 1985).
Structural break	A statistically significant shift in the parameters or variance of a time series at a particular point in time, typically associated with a major economic or political event, after which the data-generating process differs from before (Perron, 1989).
GARCH	Generalised Autoregressive Conditional Heteroskedasticity. An extension of Engle's (1982) ARCH model in which the conditional variance depends on both past squared residuals and past conditional variances, capturing the volatility clustering observed in financial data (Bollerslev, 1986).

Term	Definition
ICSS	Iterative Cumulative Sum of Squares. A sequential procedure for retrospectively detecting multiple structural breaks in the unconditional variance of a time series by analysing the centred cumulative sum of squared observations (Tiao & C., 1994).
Multicollinearity	A condition in multiple regression where two or more explanatory variables are substantially correlated with one another, inflating standard errors and making it difficult to identify the independent effect of each regressor (Haberler, 1936).
Omitted-variable bias	The bias introduced into OLS estimates when a variable that is relevant and correlated with an included regressor is excluded from the model, causing the included variable's coefficient to absorb part of the omitted variable's effect (Hausman, 1978).
Demand-side shocks	An unexpected or exogenous change in demand. In macroeconomics a demand shock means a change in aggregate demand, such as a rise or fall in autonomous consumption, investment, or exports (Coreecon, 2024).
Supply side-shocks	An unexpected or exogenous change in supply. In macroeconomics a supply shock means a change on the supply side of the economy, such as a rise or fall in oil prices or an improvement in technology (Coreecon, 2024).
Adverse Selection	A form of information asymmetry in which uninformed traders or dealers systematically face the risk of trading against a counterparty with superior private information, leading to wider bid–ask spreads as compensation (Glosten, 1985).
Cross-sectional regression	A regression estimated using observations on multiple units (e.g. firms or securities) at a single point in time, used to identify relationships between variables across entities rather than over time (MacBeth, 1973).
Endogeneity	A violation of the OLS endogeneity assumption that occurs when an explanatory variable is correlated with the error term due to simultaneity, omitted variables, or measurement error rendering least-squares estimates inconsistent (Hausman, 1978).
Volatility Regime	A persistent state of the return-generating process characterised by a distinct level of conditional variance. Markets are said to switch between regimes (e.g. low- and high-volatility states) over time, with each regime following its own stochastic process (Hamilton, 1989).

4. Literature Review

4.1. Commodity Shocks and macroeconomic transmission

When reading up on commodity dependence and equity volatility, I found that many of the dominant ideas trace back to (Hamilton, 1983), whose paper lays the foundations for much of the theory regarding transmission channels for commodity shocks.

Hamilton highlights how fluctuating energy prices propagate through markets via production costs, consumer spending and inflation expectations. This concept is later echoed across more financial-market-specific studies.

Kilian then built on this by breaking down oil shocks into supply, demand and precautionary elements, identifying distinct volatility signatures for each type of shock (Kilian, 2008). This framework is particularly useful for explaining the persistent volatility of the major oil supply disruptions in 1973 and 1979 versus the short-lived volatility of the Iraq War in 2003. Hamilton's more recent work on oil-shock modelling informed my further reading on the role of commodity dependence in financial-market stability (Baumeister, 2018).

These papers are highly reliable for the mechanisms they outline, they are formal, peer-reviewed and methodologically clear, however, most of them are not written about geopolitical equity-market volatility specifically. I therefore use them to establish how commodity shocks impact markets and then test whether those mechanisms are visible in the case-study evidence in section 6.

In applied contexts, "What do Financial Markets Think of War in Iraq?" by Leigh, Wolfers and Zitzewitz provided valuable insight into equity-market reactions to geopolitical events, particularly the role of longer-term prevailing sentiment in determining the duration of post-event volatility. For Brexit, Khurram's work on EU and UK equity-market reactions contextualised the idea of asymmetric volatility, demonstrating the way political uncertainty and market sentiment interact to magnify financial instability (Khurram, et al., 2019).

4.2 Information flow, predictability and market microstructure

On predictability and information arrival, I drew on Kyle's sequential trading model, which underpins almost all market-microstructure theory and explains many of the channels through which predictability affects market conditions and volatility (Kyle, 1985).

Fama's research on efficient capital markets was particularly useful in explaining how information gradually drips down through market makers and participants at a fundamental cost (Fama, 1970). An idea Grossman and Stiglitz then expanded on by

highlighting the inherent inefficiency within equity markets that follows from the assumption that information acquisition is costly (Grossman & Stiglitz, 1980).

The reliability of this body of work, like commodity dependence is extremely high for academic theory, however, many of the papers on market microstructure are weaker for direct application given these models assume continuous, frequent trading rather than the discrete, news-driven jumps that typically characterise geopolitical events. Therefore, I apply them with explicit acknowledgement of that disparity and combine them with Back & Baruch's work on "information arrival jumps" to capture the discontinuous case (Back, 2004).

4.3 Liquidity and market microstructure under stress

On liquidity, I turned to Amihud, who formalised the illiquidity ratio now widely used as a proxy for liquidity (Amihud, 2020). One which I often used in my own regressions. Additionally, Ramos and Righi's paper titled "Liquidity, implied volatility and tail risk: a comparison of liquidity measures" supports the view that low liquidity prolongs and intensifies volatility spikes, a finding consistent with my own regression analysis of the Trump-tariff episode (Ramos & Righi, 2020).

Furthermore, Adrian, Kiff and Shin's paper called, "Liquidity, Leverage and Regulation 10 Years After the Global Financial Crisis", informed my evaluation of liquidity as a core factor in developing markets because of the looser central-bank monetary policy they generally have (Adrian, 2018). Whereas developed markets are usually reinforced by robust liquidity backstops which reduce volatility magnitude and persistence (Leigh, 2003). The same paper supports the contrast I draw with developing markets in Section 8.

4.4 Behavioural cascades and volatility persistence

Bikhchandani, Hirshleifer and Welch provided the underpinning for the cascade-theory discussion in Section 7, in which subtle market movements lead to larger price moves as "wait-and-see investors pile in" after markets stabilise (Bikhchandani, et al., 1998). I then read up on Engle's GARCH framework which underpins the volatility-clustering and structural-break analysis used in Section 7 and 9, while Inclán and Tiao's research provided the ICSS structural-break method used to test for shifts and variation between volatility regimes (Engle, 1982) (Tiao & C., 1994).

4.5 Applied sources and limitations of the literature base

Along with academic papers I used Financial Times reporting to contextualise events and to reconstruct contemporaneous market reaction, particularly around the Russia-Ukraine conflict and US tariff disputes. Articles such as "'Investors are shocked': how

Russia's attack on Ukraine roiled markets" and "The 'Brexit effect' on UK domestic stocks" were valuable for formalising initial ideas but not used as concrete evidence for effects of the factors I cover (Wigglesworth, et al., 2022) (Bowsher, 2018).

A meaningful limitation of the literature base is that this is a relatively niche question. There is substantial academic research on volatility and on geopolitical risk separately, however, few papers address both directly. As a result, much of the literature I read was not written about my exact question. To overcome this, I read a wide variety papers and synthesised across the following strands: econometrics, international finance, market microstructure as to build an independent argument. A clear example is my use of Leigh, Wolfers and Zitzewitz's work as the paper aims to answer a different question, what financial markets thought of war in Iraq, and yet the "Saddam Security" section provided directly relevant evidence on how oil dependency shaped the financial-market response to that war (Leigh, 2003).

5. Theoretical framework

5.1 Commodity dependence: supply, demand and risk-premium channels

The fundamental purpose of equity markets is to capitalise expected future cash flows. This means that changes in supply chains and trade routes would therefore affect company valuations, largely through three connected channels. The first is supply-side shocks, which is when geopolitical events disrupt commodity supplies resulting in import-dependent economies facing immediate production constraints. Key factors of production imports like energy, metals, or agricultural products become prohibitively expensive as supply tightens, reducing future earnings expectations and placing downwards pressures on equity valuations. The uncertainty of these supply-chain recoveries is likely to amplify volatility as investors weigh up the timeline of production chain restoration. This doubt is constantly being repriced into the market causing significant movements in equity markets.

Secondly, demand-side shocks in commodity markets increase their prices and erode consumer purchasing power as the general price level rises with energy and food prices. Discretionary spending falls, service sectors contract, and retail demand decreases. Equity markets must therefore reprice firms' valuations and cash flows in response, heightening volatility as new information continually alters sentiment.

Thirdly, risk-premium and capital-flight effects cause economies to undergo fundamental changes because of high commodity dependencies. As a result, markets demand larger risk premiums on commodity-reliant economies. This is referred to as a "dependency discount" which widens as investors reassess exposure. A likely consequence of these premiums is something known as capital flight which refers to

“short-term speculative capital outflows”, driven by both shifts in domestic portfolios to safer foreign liquid assets and tighter trade-credit conditions as banks become increasingly risk-averse (Cuddington, 1986). This creates negative pressure on dependent equity markets while benefiting alternative destinations, generating volatility between national markets.

5.2 Predictability: information flow and market microstructure

Predictability is significant because markets are inherently forward-looking, constantly weighing up future expectations. If an event is anticipated, investors have time to adjust portfolios, hedge exposure, and incorporate information gradually. Contrastingly, if an event is sudden, information arrives all at once and market participants must revalue their assets under intensified uncertainty. A key difference that explains why unpredictability raises volatility, even when the actual economic damage is not yet clear.

Under rational expectations, market prices should reflect all available information. Under geopolitical stress, the informational landscape of markets is constantly shifting so the extent to which information is fully and quickly incorporated largely dictates the length of equity markets reactions, which depends on the information structure. Kyle's sequential trading model describes how an informed trader gradually releases information to noise traders and market makers, moving the price toward its true value at a steady pace as prices learn from the informed trader's activity (Kyle, 1985).

Geopolitical events frequently violate this assumption. Information is no longer released gradually but dumped on all participants simultaneously and without order. The resulting asymmetric information leads to a less efficient pricing mechanism as information no longer trickles down markets, leading to volatile prices.

Predictable events allow for what Grossman and Stiglitz call "costly information acquisition" (Grossman & Stiglitz, 1980). If markets were perfectly efficient, prices would reflect all information and there would be no incentive to gather it. But crucially, information has a real opportunity cost (time, money, effort), so prices only partially reflect information. This creates a rational incentive for informed trading and produces an "informational equilibrium" in which a predictable event is absorbed smoothly, converging to fundamental values.

Unpredictable events, by contrast, generate what Back and Baruch identify as "information arrival jumps" (Back, 2004). Sudden imbalances in information give rise to what is referred to as an adverse-selection problem in which market makers fear they are trading with someone better informed than them and widen bid-ask spreads to

compensate. This reduces liquidity and amplifies volatility, not because of fundamentals but because of a deterioration in trading conditions.

This is a direct example of how the predictability mechanism connects directly to the liquidity mechanism, a connection which is central to the ranking I defend in Section 10.

5.3 Liquidity: depth, spreads and feedback loops

Microstructure theory emphasises widening bid-ask spreads and reduced market-maker involvement as the primary cause of sharp volatility episodes. As Ramos and Righi show, the mechanism works through the share price impact of trades (Ramos & Righi, 2020). If liquidity is thin, each trade exerts greater pressure on prices because of lower market depth and activity. This can create a very damaging cycle as when market makers face greater inventory risk and adverse selection, they widen spreads out of fear that they are trading with someone better informed. This discourages further trading which results in a trading-condition disruption that raises volatility, particularly at the tails of the return distribution.

Critically, this feedback loop between higher volatility and weak liquidity prevents the market from operating efficiently and incorporating information smoothly, resulting in longer and larger price swings. It is this mechanism and crucially not the underlying news that explains why the same-sized shock can produce very different volatility outcomes in different markets and at different points in the economic supply chains.

6. Factor 1 - Commodity dependence

6.1 Main claim

Commodity dependence is most effective as an explanation for the initial magnitude of volatility, because it gives investors a direct channel through which a geopolitical shock changes expected costs, margins and inflation. However, it is a weaker explanation of persistence, because markets can adapt through substitution, hedging, and the repricing of long-run supply expectations.

6.2 Mechanism

The mechanism (set out fully in Section 5.1) operates through three main channels. The first being supply-side cost shocks which raise firms' input costs and squeeze margins. Furthermore, demand-side erosion caused by higher general prices which reduce discretionary spending and service-sector revenue can also affect markets. As well as risk-premium adjustments that widen the "dependency discount" on equity markets which are vulnerable to the affected commodity. Each channel reduces expected

future cash flows and forces investors to reprice valuations, which is realised in markets as price volatility.

6.3 Evidence: Russia-Ukraine, 2022

The clearest case of the commodity-dependence mechanism in action is the Russia-Ukraine conflict given Russia's prominence as an exporter of energy, metals, wheat and fertiliser. This means that the commodity exposure of developed economies varied sharply across European, US and Asian equity markets, providing a useful spectrum to test the relationship.

The country-by-country gas-dependency and volatility data is shown in Figure 1.

Region/Country	Russian gas dependency (%)	Main market index	Volatility response
Germany	65	DAX	Extreme
EU average	45	Euro Stoxx 50	High
United States	0	S&P 500	Moderate
Japan	8	Nikkei 225	Low-moderate

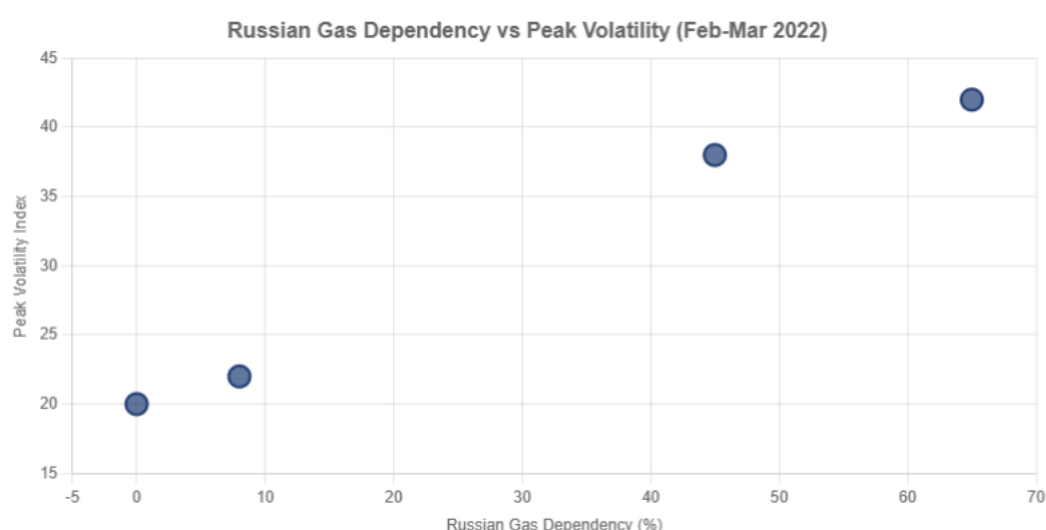


Figure 1. Russian gas dependency by country plotted against equity-market volatility response, February-March 2022. Source: IEA gas-import data; CBOE/Eurex volatility indices; author's calculations (Agency, 2023) (Louis, 2026a) (STOXX, 2026) (UK, 2026).

The cross-sectional regression yields a correlation coefficient of 0.84 between gas dependency and peak volatility response. A simple OLS regression of volatility on gas dependency for the same window returns $R^2 = 0.71$, suggesting that 71% of the cross-

sectional variation in equity-market volatility during the conflict is associated with Russian gas exposure. The fitted line is approximately:

$$\text{Volatility} = 0.36 \cdot (\text{Gas Dependency}) + 18.8$$

What this regression suggests is that for every 10-percentage point rise in commodity dependency, an economy is likely face around a 3.6-point higher volatility response. A relationship that is consistent with the theory covered in Section 5.1. The more directly an economy's firms depend on Russian commodities, the more aggressively investors are going to reprice expected cash flows thus demanding a larger risk premium for that exposure.

The volatility timeline (Figure 2) confirms this temporal pattern. The VIX moves from 28 to 37.5 on invasion day, while VSTOXX moves from 25 to 35+, with European volatility persisting at higher levels through the sanctions period and gas prices up 50–60% over the same window. This is consistent with markets penalising commodity-dependent regions both more sharply and for longer.

Date	Event	VIX (US)	VSTOXX (EU)	Gas prices
Feb 23 2022	Pre-invasion	28.0	25.0	Baseline
Feb 24 2022	Invasion day	37.5	35.0+	+50%
Mar 2022	Sanctions peak	30.0	40.0+	+60%

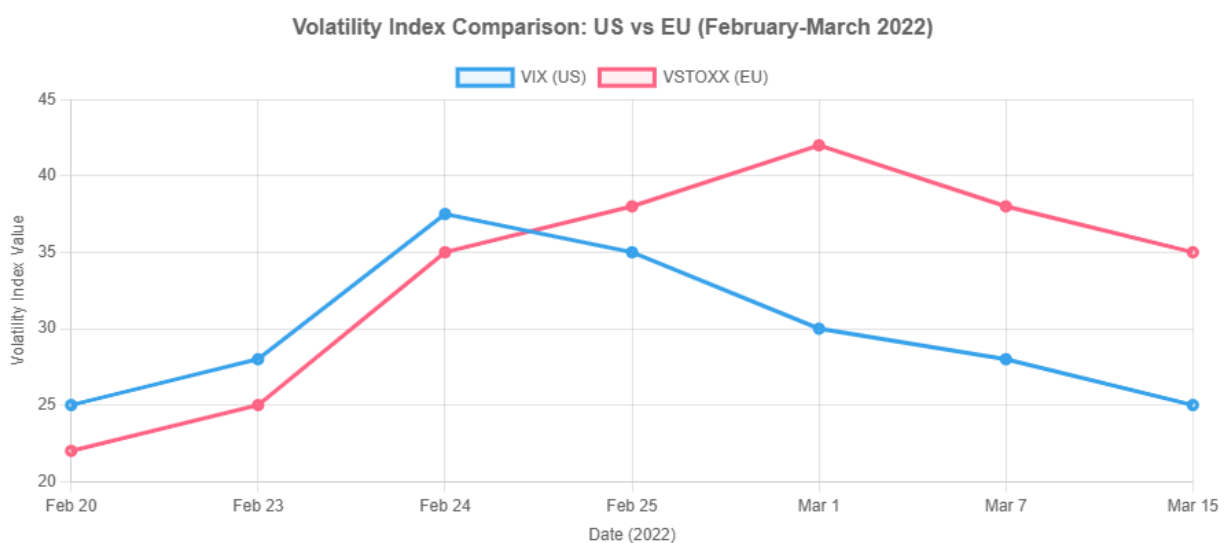


Figure 2. VIX, VSTOXX and natural-gas-price movements around the invasion of Ukraine. Source: Yahoo Finance and STOXX (UK, 2026) (STOXX, 2026).

6.4 Counter - Argument: the Iraq War, 2003

The relationship that commodity dependency raises equity-market volatility is supported by the Russia-Ukraine evidence above. I initially thought that commodity dependence would be important in explaining both magnitude and persistence given the long-lasting impact of supply chain disruptions. However, the extent to which commodity dependency is the most important factor in determining the level of volatility weakens with time. The Iraq War is the cleanest test case for that weakening. Following the US invasion in March 2003, defence, airlines and the energy sector all experienced large changes in valuation in the short term. But importantly, this volatility was short-lived, contrary to the eight-year duration of the war, and there was very little volatility in long-term oil futures as seen in Figure 3 (Leigh, 2003).

Why? While oil prices spiked on news of the conflict, investors recognised that long-run global oil supply would not be permanently impaired. From a geopolitical perspective, US involvement in Iraq had two broad possible outcomes: either the US would consolidate influence in the region and stabilise production, or it would withdraw without achieving meaningful control. In both cases, global oil flows would eventually normalise; through US-backed stability or through Iraq reverting to supplying the market alongside continued production from Iran and other producers. Investors could therefore anticipate that the disruption was temporary, and commodity dependence in this case did not translate into a lasting supply shock causing initial volatility to subside relatively quickly.

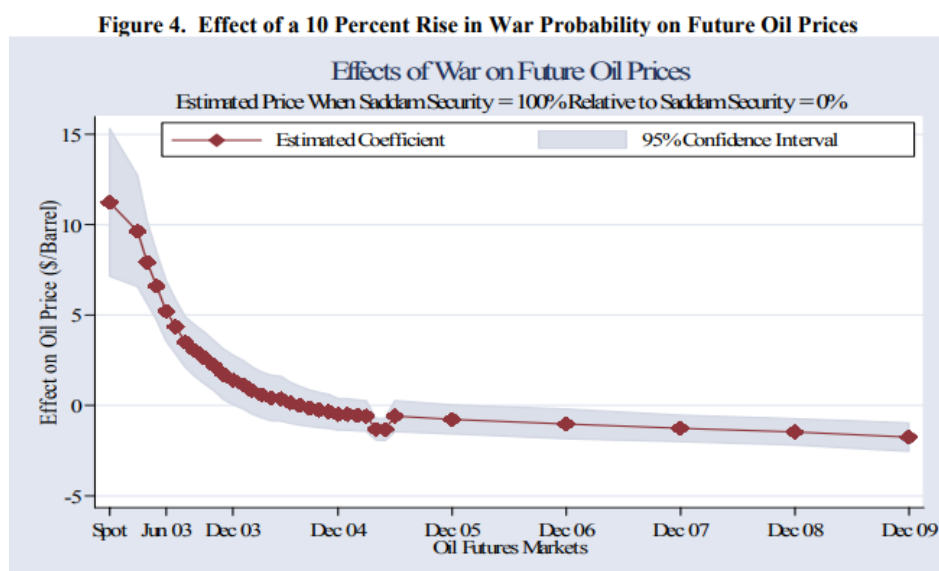


Figure 3. Long-term oil futures March 2003 - December 2009. Source: "What do Financial Markets Think of War in Iraq?" by Leigh, Wolfers and Zitzewitz (Leigh, 2003).

A second mechanism that weakens commodity dependence as a persistence factor is the development of financial instruments. Tools like options and futures contracts

enable both firms and investors to lock in commodity prices in advance. An airline can hedge fuel-cost risk; or an investor with energy-sector exposure can hedge against oil-price-driven equity declines. When this effect is compounded across investors, the initial volatility spike still occurs, but because investors are already protected, they are less likely to make panic sales and move markets violently leading to less prolonged volatility.

OPEC's decision in early 2025 to hike oil supply in anticipation of the US bombing of Iran is a recent example of the same pattern, where a commodity shock that would otherwise be expected to cause persistent equity volatility, is mitigated by producers signalling alternative supply (Hendrix, 2025).

6.5 Direct weighing of the counterargument

The Iraq evidence does not disprove commodity dependence as a factor; it narrows it. Commodity dependence is most powerful for initial amplitude and the Russia-Ukraine cross-sectional regression in Section 6.3 supports this strongly, but it is unreliable as an explanation of persistence unless the supply disruption remains physically unresolved and cannot be hedged or substituted. Even so, this short-lived volatility should not be dismissed: prolonged conflicts can permanently alter firms' cost bases, or revert to pre-war supply routes, both of which can re-trigger episodes of heightened volatility later in the conflict.

6.6 Mini-judgement

Commodity dependence ranks highly for explaining where volatility first appears and how large the first repricing is, but it does not consistently explain how long volatility lasts. It is therefore a powerful conditional driver of initial magnitude, not the most important factor overall.

7. Factor 2 - Event predictability

7.1 Main claim

Predictability is the second most important factor overall because it shapes the speed and disorder of information incorporation. Anticipated events can be priced in gradually; unanticipated events cause discrete information jumps that disrupt market microstructure. Its ranking is reinforced because predictability operates partly through liquidity which is the dominant transmission channel established in Section 8; however, it remains distinct from liquidity in that it sets the conditions under which liquidity stress arises.

Bid-Ask spreads widening in IG credit markets

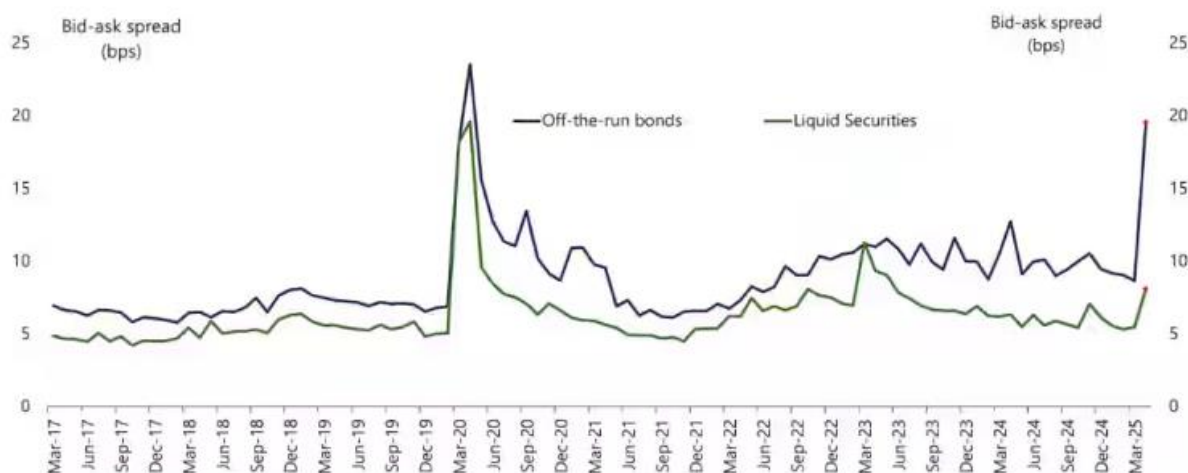


Figure 4. Bid-Ask Spreads in IG credit markets March 2017 – March 2025. Source: Journal Article titled “Bid-Ask Spreads Widening in IG Credit Markets” (Sløk, 2025).

7.2 Mechanism

Under rational expectations, predictable events are absorbed smoothly: informed traders gradually release information to market makers and noise traders, and prices converge to fundamental values along a manageable path. Unpredictable events violate this model. Information arrives all at once, market makers fear adverse selection, and bid-ask spreads widen to compensate as shown in Figure 4 below (Back, 2004). This raises volatility through a deterioration in trading conditions rather than through a change in firms’ fundamentals. Additionally, it highlights how the predictability mechanism connects directly with liquidity.

Two further mechanisms reinforce the predictability effect. The first being behavioural cascades in erratic post-shock markets which can cause the social value of information to exceed the value of private information, causing investors to mimic each other and amplify subtle moves into larger ones (Bikhchandani, et al., 1998). Cognitive limits on simultaneous information processing known as, "attention bottlenecks," make the problem worse during a sudden geopolitical event because markets cannot absorb all available information at once.

Secondly, GARCH-style volatility clustering might emerge from unpredictable events creating a "GARCH shock" in which a sudden jump in volatility feeds back into future volatility through autoregressive processes (Engle, 1982). This is the main reason why markets remain jittery for weeks after an unexpected event as poorer conditions and weaker investor confidence have essentially trained the market to be more volatile .

Structural-break theory adds the further point that some events do not just raise volatility temporarily but permanently shift the volatility regime.

7.3 Evidence: Trump-era tariffs, 2024–25

Trump's tariff policy is a useful natural experiment in unpredictable policymaking. In order to investigate this relationship, I applied the Inclán–Tiao ICSS algorithm to long-term equity-market data as to detect structural breaks in volatility (Tiao & C., 1994). I found a clear clustering of breaks around historically turbulent periods such as: the early-1980s stagflation crisis, the dot-com bubble (2000), the Global Financial Crisis (2008), and the COVID-19 pandemic (2020), which are all events that correspond to sharp declines in predictability, suggesting that markets are more susceptible to abrupt variance shifts when faced with unforeseen or poorly anticipated shocks.

When plotting predictability scores against structural-break frequency, the scatter diagram presents a strong inverse relationship ($R^2 = 0.840$, $P < 0.01$, $n = 57$) demonstrating that the more unpredictable an event, the more frequently the volatility regime changed. This is just like the "information arrival jumps" Back and Baruch (2004) describe. Reinforcing the idea that predictability functions as a stabilising force in financial markets. When investors can anticipate and model scenarios, information is gradually absorbed, consistent with Grossman and Stiglitz's framework of costly information acquisition. When events are unexpected, the variance structure undergoes discrete jumps which the ICSS algorithm effectively captures.

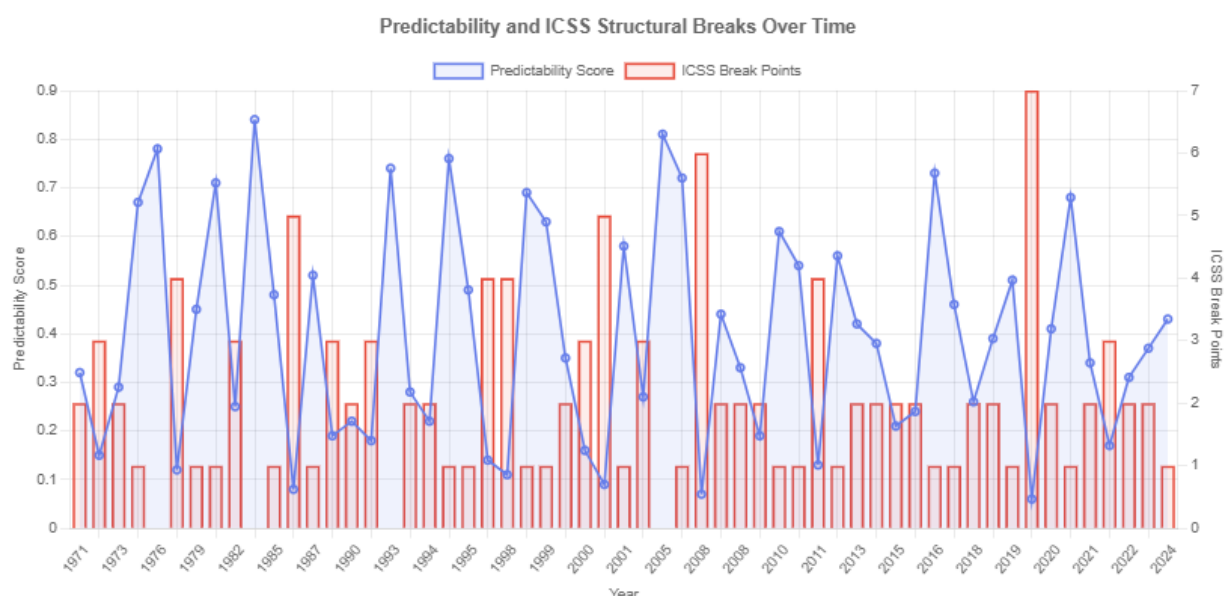


Figure 5. ICSS Structural Breaks Time Series 1971-2024. Source: Wikipedia for dates of Stock Market Crashes; CBOE VIX volatility index; author's calculations (Wikipedia, 2025) (Louis, 2026a).

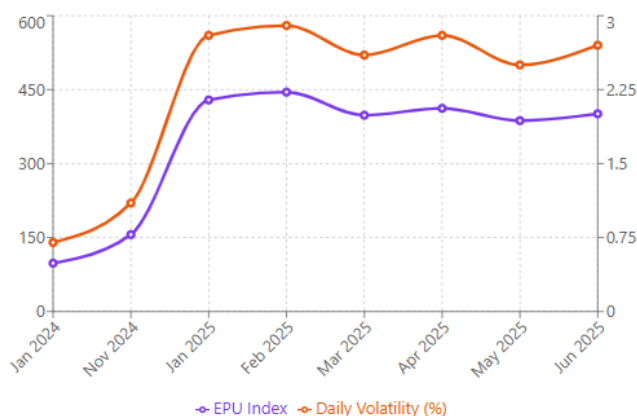
7.4 Counterargument: Brexit, 2015-2020

I chose Brexit as my counter-argument empirical investigation because it was sufficiently anticipated and yet still produced sustained equity-market volatility. Both the Trump tariff period and Brexit produced elevated economic policy uncertainty (EPU) and daily market volatility, but Brexit's rise in EPU was sharper initially while volatility settled more quickly. ICSS also detected a higher frequency of structural breaks during the Trump-tariff period than during Brexit, despite Brexit being a larger first-order political event. A naive reading would conclude that predictability matters only as a binary, considering the known versus the unknown. The Brexit case complicates that reading. Brexit was anticipated in timing but unresolved in consequence as the referendum date was known, but the implications for trade, regulation, and UK-listed firms remained uncertain for years. Predictability is therefore not simply about whether the event was foreseen, but whether the market can assign probabilities to outcomes. Where the path of an event remains uncertain even after it occurs, predictability does not eliminate volatility.

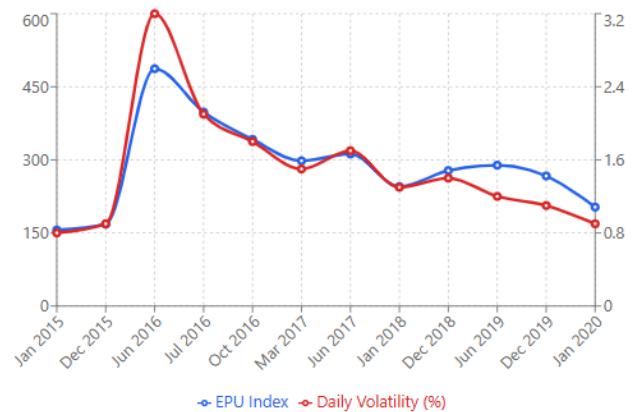
7.5 Direct weighing of the counterargument

However, I believe that the Brexit comparison supports the predictability claim rather than challenging it. Predictability is best understood as a modifier of how shocks are processed, not merely as an on/off switch. The key difference between Brexit and Trump tariffs is the inconsistency of Trump's policymaking, which rearranged fundamental trading conditions in the way Engle's GARCH framework would predict. Brexit's anticipated path produced a sharper one-off repricing without the same regime-shift behaviour, even though it remained politically prolonged. This is why predictability is the second most important factor overall but is most powerful when it interacts with liquidity stress.

Trump Tariffs: Economic Policy Uncertainty vs Market Volatility (2015-2020)



Brexit: Economic Policy Uncertainty vs Market Volatility (2015-2020)



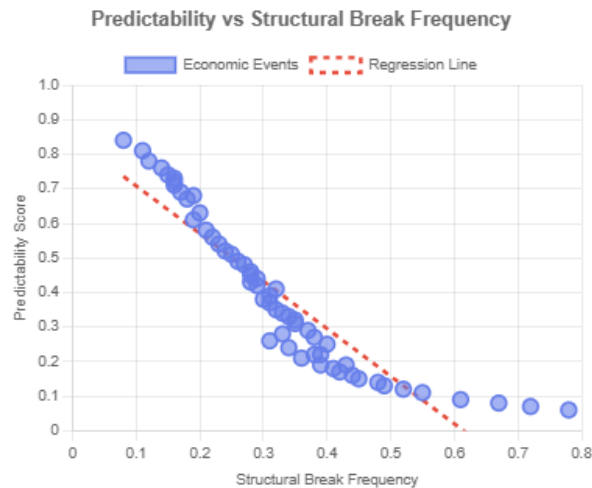


Figure 6. *Economic Policy Uncertainty Index vs Market Volatility 2015-2020 and 2024-2025. Source: Federal Reserve Bank of ST Louis for EPU index; CBOE/Eurex volatility indices; author's calculations (Louis, 2026b) (Louis, 2026a) (STOXX, 2026).*

7.6 Mini-judgement

Predictability is best understood as a modifying factor rather than the primary driver of equity market volatility. It shapes the speed and disorder of repricing, but it becomes most consequential when combined with liquidity stress or high commodity exposure.

8. Factor 3 - Liquidity

8.1 Main claim

Liquidity is the most important overall factor in determining the level of volatility a geopolitical event causes in global equity markets, because it determines the extent to which the market can assimilate new information without disorderly price movements. Commodity dependence and predictability explain why investors want to trade; liquidity explains what happens when they all try to trade at once. If market depth is strong, the shock is absorbed through normal repricing. If liquidity is weak, that same shock can produce a self-reinforcing volatility spiral.

8.2 Mechanism

The mechanism operates through market depth, bid-ask spreads, and inventory risk. When liquidity falls, market makers widen spreads because holding inventory becomes riskier and adverse selection rises. Investors face higher trading costs, which discourages stabilising trades and increases forced selling pressure. Price changes become larger for a given trade size, so volatility rises even if the fundamental shock has not worsened. This makes liquidity both an independent factor and a transmission channel for predictability and commodity exposure.

As Ramos and Righi show, the mechanism produces effects particularly at the tails of the return distribution causing the extreme losses and gains that drive the volatility headlines (Ramos & Righi, 2020). Critically, the feedback loop between higher volatility and weak liquidity prevents the market from incorporating information smoothly, leading to more persistent and larger price swings.

8.3 Evidence: April 2025 tariff episode and the liquidity regression

Findings from April 2025, the peak of Trump's tariff measures, support the liquidity hypothesis directly. Spreads widened dramatically in the immediate aftermath of the April 2nd tariff announcement with investment-grade credit spreads rising to near-COVID levels (Sløk, n.d.). Most importantly, volatility spikes coincided not with the peak of unpredictability but with the most severe liquidity stress.

To test this empirically, I ran a cross-sectional regression on S&P 500 stocks during April 2025. Two models were estimated: one using liquidity variables (bid-ask spreads and the Amihud illiquidity ratio), the other using policy-predictability metrics (EPU and tariff-sentiment scores).

Model	Variables used	R ²	Correlation with volatility
Liquidity model	Bid-ask spread, Amihud illiquidity ratio	0.572	0.76
Policy model	EPU, tariff sentiment score	0.073	0.27

Figure 7. Cross-sectional regression of S&P 500 stock-level volatility on liquidity vs policy-uncertainty proxies, April 2025. Source: author's calculations using LSEG/WRDS bid-ask data, EPU index and stock-level realised volatility (Louis, 2026a) (Louis, 2026b) (Baker, 2016; LSEG, 2026; WRDS, 2026).

Liquidity variables explained 57.2% of cross-sectional variation in stock-level volatility compared to only 7.3% for policy variables. Correlation with volatility was 0.76 for liquidity and 0.27 for policy. This supports, though importantly is not formal proof of, the conclusion that liquidity was the dominant transmission channel during this episode. Ramos and Righi (2020) report similar findings in a different setting: low liquidity both amplifies and prolongs volatility by restricting the market's ability to absorb shocks.

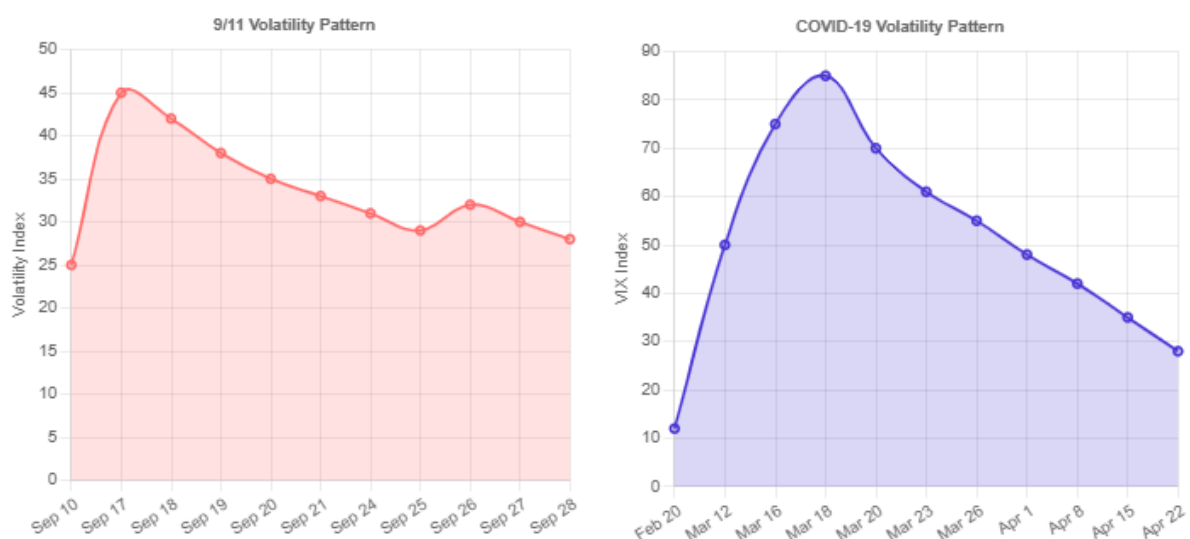
8.4 Counterargument: Central Bank Policy

There are two key counterarguments to this liquidity claim. First, the endogeneity counter suggesting that liquidity may not be entirely independent of predictability. In April 2025, the deterioration in liquidity arguably stemmed from investor reaction to

Trump's combative and abrupt policymaking. Uncertainty drove down investor confidence and increased the cost of information acquisition, which widened spreads and caused investors to pull back. On this reading, liquidity is a manifestation of predictability, not an alternative to it. In my opinion, this strengthens the role of liquidity rather than weakening it because it demonstrates how other such factors can operate through it.

Secondly, the central-bank counter which highlights how liquidity may not be decisive in developed markets where central banks act as backstops. Comparing the post-9/11 and COVID-19 volatility episodes illustrates this. After the Global Financial Crisis, fundamental market conditions and core principles of monetary policy were almost uprooted, with the Fed, ECB and Bank of Japan substantially expanding balance sheets and flooding markets with liquidity. The Fed's balance sheet expanded by over 100% during COVID-19 versus a modest 17% increase after 9/11. The result was that although the initial COVID-19 volatility spike was much sharper (VIX above 80), it was also much less persistent, while the more modest 9/11 volatility persisted for months as seen in the figures below.

Practitioner context (Appendix A): a former Neuberger Berman professional whose role had involved working with official institutions including central banks emphasised that since the Global Financial Crisis, central banks have become more willing to act as liquidity providers during periods of severe market stress. Their attitudes have fundamentally shifted, which is made clear in Figure 7 below as Central Bank balance sheets have rapidly expanded since the GFC. This conversation is used as supporting practitioner context, not as proof of any direct link.



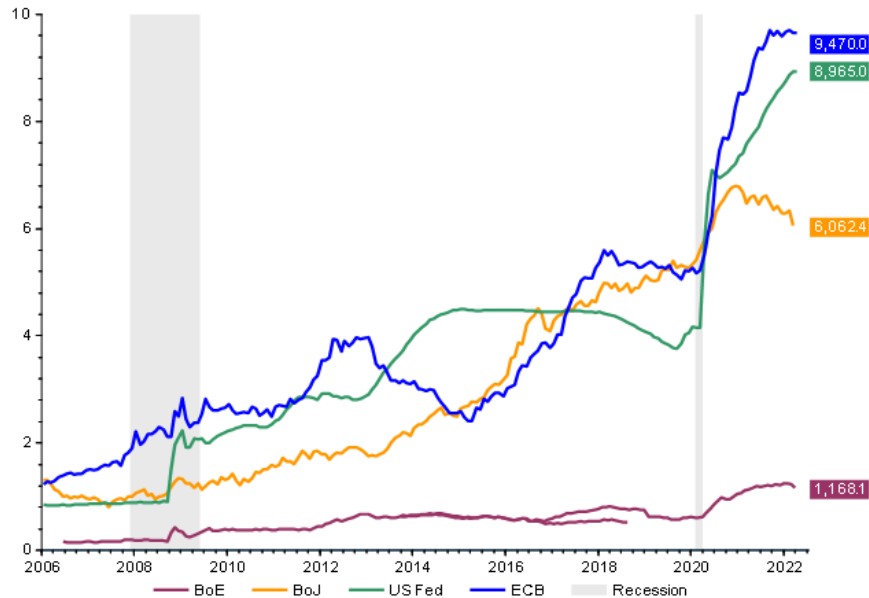


Figure 8. S&P 500 stock-level volatility time series, September 2011 & February – March 2020. Central Bank Balance sheets 2006 - 2022. Source: author's calculations using CBOE volatility index and OMFIF research (UK, 2026) (Williams, 2022).

8.5 Direct weighing of the counterarguments

Crucially, both counter-arguments refine the liquidity claim rather than displacing it. On endogeneity, the fact that liquidity stress was caused by Trump's unpredictable policymaking does not change the fact that liquidity stress was the channel through which that unpredictability became persistent volatility. The 5.7 times larger R^2 for the liquidity model than the policy model in Section 8.3 suggests that cross-sectional regression is explained far better by liquidity conditions rather than policy-related factors. Despite policy news likely triggering the initial shock. The weakening in liquidity conditions instead of the news is what resulted in the differences in stock market volatility variation. Liquidity is therefore the most significant transmission mechanism, even if it is not always the root cause, however, liquidity is also not just an indirect transmission mechanism. Following conversations with a loan structuring professional (Appendix A – Personal Communication 2), I was made aware of how tighter liquidity conditions dry up funding liquidity for corporates translating into lower investment and earnings expectations for firms which alter valuations.

Concerning central-bank backstops, developed-market policy intervention has reduced the visible impact of liquidity stress, but it has done so by reinforcing the importance of liquidity as the pivotal factor. The evolution of central-bank policy, global coordination, the development of QE, forward guidance and yield-curve control, and the willingness to expand balance sheets at greater scale (an approximate \$5tn increase in the Fed balance sheet during COVID-19) has not removed liquidity from the volatility equation.

It has changed who provides it. The expectation of central-bank support has itself become part of the volatility mechanism given that when investors believe a backstop will appear, the volatility associated with a given shock is smaller.

In developing markets, the picture is different and reinforces the same point. In E7 economies such as Brazil, India and South Africa, advanced monetary tools and central-bank depth are limited, leaving these markets more vulnerable to persistent volatility. Demonstrating how global financial cycles driven largely by US monetary policy affect emerging-market capital outflows and asset prices independently of domestic conditions (Rey, 2015). The 2013 "Taper Tantrum" is the canonical illustration of this. Hints from then-Fed Chair Ben Bernanke about tapering QE produced a deep bond sell-off, heightened cross-asset volatility and capital flight from emerging markets resulting in EM currencies plummeting as well as equity drawdowns and volatility spikes following as foreign investors de-risked. EM central banks, lacking the capacity to expand balance sheets at the scale of the Fed or ECB, responded with defensive rate hikes and reserve interventions that tightened domestic conditions further, amplifying volatility.

8.6 Mini-judgement

Liquidity is the most important overall factor because both its absence and its credible provision shape whether geopolitical-shock volatility persists. The endogeneity counter sharpens this as even where liquidity stress is caused by unpredictability, it is the liquidity stress that converts unpredictability into a persistent volatility regime. Additionally, the central-bank counter does not weaken the claim either. Rather it shows that liquidity is so important that policy attention has been redirected to defending it.

9. Comparative judgement

Before stating the final conclusion, the table below summarises how each factor performs against the four criteria for "importance" set out in Section 1.1.

Factor	Initial magnitude	Persistence	Independence	Evidence quality	Overall judgement
Commodity dependence	High where the event affects essential inputs (energy, food, metals)	Medium-low unless physical supply disruption is unresolved and difficult to hedge against.	Moderate as it interacts with inflation and policy response very closely.	Strong for most geopolitical events such as Russia-Ukraine given commodities are	Conditional amplifier of initial volatility.

Factor	Initial magnitude	Persistence	Independence	Evidence quality	Overall judgement
				pivotal to geopolitics.	
Predictability	Moderate given sudden events cause jumps, but size depends on exposure to the event.	High because poor forecast ability creates repeated repricing and regime shifts.	High, but partly works through liquidity and sentiment.	Strong theoretically however mitigated empirically because of proxy limits	Significant driver of volatility persistence.
Liquidity	High given thin markets convert shocks into outsized price moves.	High when spreads stay wide and depth remains weak.	Medium because it is often caused by uncertainty rather than independent of it.	Strong across nearly all instances given liquidity is fundamental part of every equity market	Strongest immediate transmission mechanism and most important overall
Central-bank response	Reduces magnitude after the initial shock	High in developed markets but absent or weak in EM.	Not an original factor rather it is driven by liquidity and market conditions.	Good qualitative support; needs cautious use	Important moderator, especially post-GFC

Figure 9. *Comparative judgement matrix. Each factor is scored against the four criteria for "importance" set out in Section 1.1: initial magnitude, persistence, independence, and evidence quality.*

Read column-by-column, liquidity is the only factor that scores at or near the top on initial magnitude, persistence, AND transmission power. Predictability scores top on persistence but works partly through liquidity. Commodity dependence scores top on initial magnitude in commodity-exposed events but does not generalise. The matrix supports the ranked judgement defended throughout: liquidity first, predictability second, commodity dependence third.

10. Methodological limitations

10.1 Limits of the Russia-Ukraine cross-section

Choosing Russia-Ukraine as the main commodity-dependence case study had clear advantages but also a small-sample problem. The advantages were that it gave me a clear independent-variable gradient (Germany 65% → EU 45% → Japan 8% → US 0%)

which made it possible to test the hypothesis across thresholds rather than just high-vs-low; the four economies were of similar levels of development, reducing one source of confounding that would weaken the reliability of my results. Additionally, Russia's commodity export mix (gas, wheat, fertiliser, metals) allowed me to test whether the relationship held across multiple commodity channels rather than a single one, something an Iraq-style oil-only case could not do.

However, the key limitation to this case study was that it occurred during a period of heightened macroeconomic instability and turbulence. The same period saw COVID recovery, global inflation concerns and significant changes in monetary policy, all of which independently affect equity-market volatility. A single-variable regression therefore overstates the effect of the tested variable because it does not include enough controls. With only four data points the cross-section is also too small to reach formal statistical significance at the 5% level. However, when this is combined with the high temporal clarity (volatility spiked almost immediately after the war began) and consistency with the theory in Section 5.1, the conclusion remains defensible, but it should be read as supporting evidence rather than as standalone proof.

10.2 Multicollinearity in the multivariate regression

To test what happens when all three factors are estimated together, I ran the multivariate cross-sectional model covered in Section 2.3:

$$\text{Model: } \Delta Vol = \alpha + \beta_1 \cdot \text{Dependence} + \beta_2 \cdot \text{Predictability} + \beta_3 \cdot \text{Liquidity} + \Gamma' \cdot \text{Controls} + \varepsilon$$

$$\text{Estimated: } \Delta Vol = 34.379 + 0.090 \cdot \text{Dependence} - 0.133 \cdot \text{Predictability} - 32.028 \cdot \text{Liquidity}$$

$$R^2 = 0.435 \quad n = 47 \quad \text{Individual correlations: Dep } 0.465 \cdot \text{Pred } -0.647 \cdot \text{Liq } -0.834$$

There are three key observations from this result. The first being that the explanatory power ($R^2 = 0.435$) of my model is limited in strength. Despite this, I see this as informative rather than disappointing because it confirms to me that no single or combination of factors can fully explain volatility. A result that directly supports the case for a layered ranking rather than a single-factor explanation. Secondly, the much larger absolute coefficient on liquidity (-32.028) and its much higher individual correlation (-0.834) are consistent with the Section 8 claim that liquidity is the dominant transmission channel. Thirdly, the strong negative correlation between liquidity and predictability is itself a multicollinearity warning as when variables are correlated, the regression is unable to directly attribute their effects to volatility, which is precisely the empirical signature of the Section 8.4 endogeneity argument. I believe the right interpretation is not to dismiss the model but to use it to motivate the qualitative ranking.

Looking back, it would have been more mathematically clean to use the multivariate regression as the primary analysis from the beginning and to then discuss multicollinearity as part my evaluation instead of using single-factor regressions first. This is an important methodological lesson I would carry forward to any follow-up work at university level.

10.3 Other methodological limits

Another methodological limitation I faced was that the relationship between each factor and volatility may not be linear. Instead, a polynomial functional form, or a time-lag structure, could plausibly fit better, and neither has been tested here. Furthermore, the proxies are imperfect, national energy dependence is not the same as listed-equity exposure, EPU is a noisy proxy for predictability, and the Amihud ratio depends on volume measurement. Each is the best available proxy but each leaves room for measurement error. Finally, ICSS structural-break detection is sensitive to the threshold parameter meaning not every detected break is a geopolitical event. A higher-accuracy model is achievable but lies beyond the scope of an A-level EPQ as it would belong to undergraduate-level econometrics work.

11. Conclusion

This dissertation concludes that liquidity is the most important overall factor in determining the level of volatility a geopolitical event causes in global equity markets. Predictability is the second most important factor, and commodity dependence is the third. Each factor governs a different stage of the volatility process, but I believe that is what justifies a ranked rather than a flat answer.

Commodity dependence is critical for explaining initial magnitude where an event threatens core inputs. The Russia-Ukraine evidence ($R^2 = 0.71$, correlation 0.84) shows that European equity markets responded more sharply than less-exposed US and Japanese markets, exactly as the supply-shock and risk-premium mechanisms in Section 5.1 would predict. But the Iraq War shows that commodity dependence does not necessarily produce persistent volatility when investors expect supply stability or when the shock can be hedged through derivatives. Commodity dependence is therefore a powerful but conditional driver, ranked third overall.

Predictability is the deepest determinant of how long volatility persists. Where event paths cannot be modelled with confidence, repeated repricing and regime shifts follow, which are captured statistically by the structural breaks ICSS detects in the Trump-tariff window. Conversely, Brexit shows that predictability is not binary given that an anticipated event with unresolved consequences still produces volatility. Predictability

is therefore the second most important factor, sharpest when it interacts with the liquidity channel.

Liquidity is the most important factor overall because it determines whether the shock becomes a temporary repricing event or a sustained volatility regime. The April 2025 cross-sectional regression (R^2 of 0.572 for liquidity vs 0.073 for policy proxies) makes the case directly. Even where liquidity stress is endogenous to the underlying news (the strongest counterargument), it is the liquidity stress itself that translates that news into persistent volatility. Moreover, the post-GFC evolution of central-bank backstops does not weaken this conclusion but strongly confirms it, by showing that policy attention has been redirected to defending the very channel through which volatility transmits.

The ranking is not absolute across every market. In commodity-importing economies facing direct supply disruption, commodity dependence may dominate the initial response. In developing markets with weaker policy credibility, liquidity becomes even more important because capital flight and lower market depth amplify volatility. In developed markets, central-bank backstops can reduce the persistence of liquidity-driven volatility, but this reinforces rather than weakens the importance of liquidity, because the expectation of support has become part of the volatility mechanism itself.

Overall, I believe that the strongest summary answer to the research question is therefore this, exposure determines vulnerability, unpredictability destabilises expectations, and liquidity dictates how violently that instability appears in prices. To conclude, I have found that the most important determinant of geopolitical volatility is not the shock itself but the market structure through which the shock is transmitted suggesting liquidity is the most important of the three factors.

12. Wider implications

12.1 For investors

Implications for investors are that geopolitical risk should not be assessed solely by event severity. The right questions are mechanism questions such as whether the event change cash flows through commodity exposure? Was the event anticipated, and if so, are the consequences modellable? Is market liquidity sufficient to absorb trading pressure? On the evidence in this dissertation, hedging commodity exposure addresses initial magnitude in the short term, while monitoring liquidity conditions and policy transparency is the better guide to whether volatility will persist.

12.2 For policymakers

The implication regarding policy is that clear communication and credible liquidity provision are the two most powerful tools for moderating volatility, even when the

underlying geopolitical event cannot be controlled. It also raises a longer-term concern because with central banks operating trillion-dollar balance sheets, markets have become dependent on central-bank liquidity, which is hard to unwind without destabilising equity markets. This dependency challenges long-term market stability and demands that central banks develop credible policies for adjusting their volatility response over time.

12.3 For further research

Two questions arose from this project that would be valuable to take further. The first is whether policy responses to geopolitical events are themselves more important in determining equity-market volatility than the existing pre-event factors (commodity dependence, predictability, liquidity). I have only briefly explored the extent to which these factors are mitigated by central-bank and government responses. A more direct study could refine the volatility model and might be used to forecast general financial-market behaviour during shocks.

The second is to use a single multivariate regression model with many more factors and many more events. Even at a time of heightened geopolitical uncertainty, gathering enough data points would mean dating back decades into a period where the relative importance of the factors may itself have changed because of an inconsistent macroeconomic regime, and where data is much harder to find. The same project at undergraduate level, with access to longer datasets and more advanced econometrics, would be a natural next step.

13. Evaluation

I am pleased with the project overall, particularly with my progress in learning econometrics. With the help of my supervisor, I was able to learn about the advantages and disadvantages of different regression approaches and implement them into my EPQ, while being explicit about the drawbacks of each method (now consolidated in Section 10). The use of over twenty peer-reviewed academic papers reduces the risk of factual inaccuracy or weak expert backing, and consultation with industry experts (Appendix A) supports the conclusions further. The empirical evidence I gathered and processed independently in Excel was central to the argument.

I thought the hardest part of my EPQ was the early drafting and building a clear argument. It was particularly difficult to balance the complexity of my question with clarity and conciseness. I found that the more I read the surrounding literature, the easier it became to skim less relevant papers until I encountered directly useful sections. Furthermore, as advised by my supervisor, reading the seminal papers in the surrounding fields gave me much greater clarity on what I was looking for in subsequent

reading. With hindsight, I should have begun the data-gathering phase earlier as I underestimated both the volume of data I would need and the level of organisation I would need in order to analyse it in Excel, which put me behind early-year deadlines that I later caught up on.

My research process, despite running longer than expected, was both productive and rewarding. Yahoo Finance, the IMF and JSTOR provided most of the resources I needed online with little trouble. Books from the library and regular meetings with my supervisor were the main route to expanding my knowledge of econometrics, particularly when I was working through unfamiliar formulas and concepts.

My initial research question was too vague making it difficult to begin tackling my EPQ. Narrowing the scope from "financial markets" to "global equity markets", and committing to specific case studies gave me a clear sense of direction, enabling me to tackle my question methodically. The final question invited genuine debate and made the comparative ranking in Section 9 possible. Once I had narrowed it down, the path to a coherent dissertation became much clearer and I understood what I was aiming to achieve.

If I were to write a similar dissertation again, I would focus harder on the mathematical accuracy of my methods from the beginning. I am proud of both the depth and range of academic theory I covered and applied to real-world scenarios. The remaining room for improvement lies primarily in the mathematical investigation and the presentation of empirical findings. Overall, I am pleased with the project, and the transferable skills it developed including research, referencing, planning, and evaluation. These are all skills that will be directly useful at university.

Appendix A — Interview record

Personal communication 1

Field	Detail
Format	Informal in-person conversation
Date	14 th May 2025
Role of interviewee	Managing Director and Head of Official Institutions, formerly at Neuberger Berman.
Themes discussed	Post-2008 evolution of central-bank willingness to provide liquidity during episodes of severe market stress, covering the role of Fed/ECB/BoJ balance-sheet expansion in dampening volatility persistence and the structural

Field	Detail
	vulnerability of developing economies with weaker central-bank monetary tooling.
Key insight used in dissertation	The Global Financial Crisis was a catalyst for a structural shift in central-bank attitudes to volatility management. However, importantly developing economies remain disproportionately exposed to liquidity-driven volatility because they cannot replicate developed-market backstops
Where used in dissertation	Section 8.4 (counterargument on central-bank backstops), used as supporting practitioner context, not as proof of causality.
Reliability limitation	I have made note of the subjective nature of this interview and the fact that it is not formally recorded or statistically replicable. I am also aware of the potential bias arising from his professional background.

Personal communication 2

Field	Detail
Format	Informal in-person conversation
Date	14 th May 2025
Role of interviewee	Finance-sector professional with specific experience in Strategic corporate finance and Loan Structuring.
Themes discussed	How geopolitical shocks not only tighten stock liquidity but funding liquidity. As credit conditions tighten costs for firms rise and investment opportunities dry up. Liquidity constrains real-time investor decisions and affects equity prices through fundamental mechanisms.
Key insight used in dissertation	Tight liquidity conditions don't just affect equity prices indirectly. Its role as a transmission mechanism is even more important than I originally anticipated given it can tighten funding liquidity for corporates, feeding into lower investment and earnings expectations.
Where used in dissertation	§8.5 (developing-market argument) — qualitative supporting evidence only
Reliability limitation	Subjective; informal; not replicable; potential framing effect from interviewee's professional context

Appendix B — Source reliability table

Source category	Used for	Reliability strength	Limitation or bias	How used here
Peer-reviewed theory (Hamilton, Kilian, Fama, Grossman & Stiglitz, Kyle, Engle etc).	Mechanisms behind commodity shocks, information efficiency, market microstructure and volatility clustering.	Methodology explicit; academically reviewed.	Some papers are old or not written specifically about geopolitical equity-market volatility.	Used to establish mechanisms and initial understanding. I then tested these mechanism against the case studies in Sections 6,7 and 8.
Market data (CBOE, Eurex, Yahoo Finance, LSEG, WRDS, Macrotrends)	Volatility indices, prices, spreads, trading volume and index movements.	Primary or widely used datasets with replicable time series.	Different vendors may calculate series' differently; however, I ensured I remained consistent when using volatility indices. Additionally, event windows can be arbitrary.	Exact window, proxy and calculation stated; major values cross-checked across vendors.
Macro data (IEA, IMF, World Bank, ECB, Fed)	Commodity dependence, central-bank balance sheets, macro context.	Institutional sources with transparent methods.	National aggregates do not always map onto equity-index exposure.	Proxy mismatch flagged each time (e.g. Russian gas dependency ≠ listed-company exposure)
Financial journalism (Financial Times)	Event chronology and providing me with contemporaneous market interpretation.	Editorially curated, and often fact checked. Often contextualised events and theory for me.	Not peer-reviewed which increases the likelihood that it may reflect publication framing or personal bias.	Used only for context and chronology, not as evidence of causality or academic theory.

Source category	Used for	Reliability strength	Limitation or bias	How used here
Industry interviews (Appendix A)	Practitioner interpretation of central-bank backstops and liquidity mechanisms.	I found perspective was not always captured in academic literature. I was also able to ask probe and ask questions	Subjective, not replicable and potentially subject to professional background bias	Used as supporting context in Section 8.4 & 8.5 only; not used to establish causal claims
Original Excel regressions and figures	Cross-sectional and time-series tests of the proposed mechanisms	Original synthesis using transparent data sources	Small samples (4-country Russia-Ukraine cross-section). Also limited by proxy and linearity assumptions.	Treated as supporting evidence for the qualitative ranking, not as standalone proof.

Bibliography:

Adrian, J. K. a. H. S. S., 2018. Liquidity, Leverage and Regulation 10 Years After the Global Financial Crisis. *Annual Review of Financial Economics*.

Agency, I. E., 2023. *Reliance on Russian Fossil Fuels in OECD and EU Countries*. [Online]

Available at: <https://www.iea.org/data-and-statistics/data-product/reliance-on-russian-fossil-fuels-in-oecd-and-eu-countries>

[Accessed May 2025].

Amihud, N. & J., 2020. Illiquidity and Stock Returns II: Cross-section and Time-series Effects. *The Review of Financial Studies*.

Andersen, T. B. F. X. D. P. L., 2003. Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, Paul Labys. *Econometrica*, 71(2), pp. 579-625.

Back, S. B., 2004. Information in Securities Markets: Kyle Meets Glosten and Milgrom. *Econometrica*, 72(2).

Bai, J. & Perron, P., 1998. Estimating and Testing Linear Models with Multiple Structural Changes. *Econometrica*,.

Baker, N. B. ,. S. J. D., 2016. Measuring Economic Policy Uncertainty. *The Quarterly Journal of Economics*, 131(4).

Baumeister, J. D. H. & C. J., 2018. Revisiting the Role of Oil Supply and Demand Shocks. *National Bureau of Economic Research*.

Bikhchandani, Hirshleifer, D. & Welch, I., 1998. Learning from the Behavior of Others: Conformity, Fads, and Informational Cascades. *The Journal of Economic Perspectives*, Issue American Economic Association.

Bollerslev, T., 1986. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), pp. 307-327.

Bowsher, E., 2018. *The 'Brexit effect' on UK domestic stocks*. [Online]
Available at: <https://www.ft.com/content/6814223c-faaf-11e7-9b32-d7d59aace167>
[Accessed 2025].

Caldara, D. a. M. I., 2022. Measuring Geopolitical Risk. *American Economic Review*, 112(4), pp. 1194-1225.

Campbell, J., Lo, A. & MacKinlay, A., 1997. *The Econometrics of Financial Markets*. Princeton University Press.

Contessi, P. D. P. & Silvio, 2021. The international spread of COVID-19 stock market collapses. *National Library of Medicine*.

Coreecon, 2024. *4.8 The business cycle model: Demand and supply shocks, and inflation expectations*. [Online]
Available at: <https://books.core-econ.org/the-economy/macroeconomics/04-inflation-and-employment-08-business-cycle-model-shocks-inflation-expectations.html>
[Accessed 2025].

Cuddington, J. T., 1986. Capital Flight: Estimates, Issues,. *PRINCETON STUDIES IN INTERNATIONAL FINANCE*.

Eckblad, D. B. & Marshall, n.d. *Stock Market Crash of 1987*. [Online]
Available at: <https://www.federalreservehistory.org/essays/stock-market-crash-of-1987>

Engle, R. F., 1982. Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation. *Econometrica*, Volume 50.

Fama, E., 1970. Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance*, Volume 25.

Fidelity, 2026. *Understanding stock market volatility and how it could help you*. [Online]
Available at: <https://www.fidelity.com.sg/beginners/your-guide-to-stock-investing/understanding-stock-market-volatility-and-how-it-could-help-you>

Glosten, P. M., 1985. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), pp. 71-100.

Grossman, S. & Stiglitz, J., 1980. On the Impossibility of Informationally Efficient Markets. *The American Economic Review*.

- Haberler, G., 1936. Some Reflections on the Present Situation of Business Cycle Theory. *The Review of Economics and Statistics*, 18(1), pp. 1-7.
- Hamilton, J., 1983. Oil and the Macroeconomy since World War 2. *Journal Of Political Economy*.
- Hamilton, J. D., 1989. A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle. *Econometrica*, 57(2), pp. 357-384.
- Hansen, B., 2001. The New Econometrics of Structural Change.. *Journal of Economic Perspectives*.
- Hausman, J. A., 1978. Specification Tests in Econometrics. *Econometrica*, 46(6), pp. 1251-1271.
- Hendrix, C. S., 2025. Why oil markets shrugged at the US bombing of Iran. *Realtime Economics*.
- Ho, H. R. S., 1981. Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics*, 9(1), pp. 47-73.
- Ita, D.-A., 2025. *Understanding Equity Markets: Definition, Function, and Global Examples*. [Online]
Available at: <https://www.investopedia.com/terms/e/equitymarket.asp>
[Accessed 2025].
- Kahneman, D. & T. A., 1979. Prospect Theory: An Analysis of Decision under Risk. *Econometrica*,.
- Khurram, S., Ghulame, R., M.A.P.E., L. & Tariq, B., 2019. UK's stock market reaction to Brexit process: A tale of two halves. *Economic Modelling*, Volume 80.
- Kilian, L., 2008. Exogenous Oil Supply Shocks: How Big Are They and How Much Do They Matter for the U.S. Economy?. *The Review of Economics and Statistics*.
- Köstlmeier, S., 2021. *ICSS: ICSS Algorithm by Inclan/Tiao (1994)*. [Online]
Available at: <https://cran.r-project.org/web/packages/ICSS/index.html>
- Kyle, A., 1985. Continuous Auctions and Insider Trading. *Econometrica*, 53(6), p. 21.
- Leigh, A. W. J. & Z. E., 2003. What do Financial Markets Think of War in Iraq?. *National Bureau of Economic Research*.
- Louis, F. R. B. o. S., 2026a. *CBOE Volatility Index: VIX (VIXCLS)*. [Online]
Available at: <https://fred.stlouisfed.org/series/VIXCLS>
[Accessed April 2025].
- Louis, F. R. B. o. S., 2026b. *Economic Policy Uncertainty Index for United States (USEPUINDXD)*. [Online]

Available at: <https://fred.stlouisfed.org/series/USEPUINDXD>
[Accessed 2025].

LSEG, 2026. *LSEG Data & Analytics*. [Online]
Available at: <https://www.lseg.com/en/data-analytics>
[Accessed 2025].

MacBeth, E. F. F. a. J. D., 1973. Risk, Return, and Equilibrium: Empirical Tests. *Journal of Political Economy*, 81(3), pp. 607-636.

Management, G. A., 2021. *Understanding and Using VSTOXX® Futures*. [Online]
Available at: <https://www.deutsche-boerse.com/resource/blob/2404636/ce2206373a6cd86799589b0f3c322806/data/vstox-01-2021.pdf>
[Accessed 2025].

Nations, U., 2023. *The State of Commodity Dependence 2023*. [Online]
Available at: <https://unctad.org/publication/state-commodity-dependence-2023>
[Accessed 2025].

Perron, P., 1989. The Great Crash, the Oil Price Shock, and the Unit Root Hypothesis. *Econometrica*, 57(6), pp. 1361-1401.

Ramos, H. P. & Righi, M. B., 2020. Liquidity, implied volatility and tail risk: A comparison of liquidity measures. *International Review of Financial Analysis*, Volume 69.

Rey, H., 2015. Dilemma not Trilemma: The Global Financial Cycle and Monetary Policy Independence. *National Bureau of Economic Research*.

Shiller, R., 1981. Do Stock Prices Move Too Much to be Justified by Subsequent Changes in Dividends?. *American Economic Review*.

Sløk, T., 2025. *Bid-Ask Spreads Widening in IG Credit Markets*. [Online]
Available at: <https://www.apolloacademy.com/bid-ask-spreads-widening-in-ig-credit-markets/>
[Accessed 2025].

STOXX, 2026. *EURO STOXX 50® Volatility (VSTOXX®)*. [Online]
Available at: <https://www.stoxx.com/data-index-details?symbol=V2TX>
[Accessed May 2025].

Tiao, C. I. & C., G., 1994. Use of Cumulative Sums of Squares for Retrospective Detection of Changes of Variance. *Journal of the American Statistical Association*, Volume 89.

UK, Y. F., 2026. *CBOE Volatility Index (^VIX)*. [Online]
Available at: <https://uk.finance.yahoo.com/quote/%5EVIX/>
[Accessed 2025].

Wigglesworth, R. et al., 2022. *'Investors are shocked': how Russia's attack on Ukraine roiled markets*. [Online]
Available at: <https://www.ft.com/content/828e786f-fc62-47db-920c-f683211ca853>
[Accessed 2025].

Wikipedia, 2025. *List of Stock Market Crashes and Bear Markets*. [Online]
Available at:
https://en.wikipedia.org/wiki/List_of_stock_market_crashes_and_bear_markets
[Accessed 02 11 2025].

Williams, N., 2022. *Central banks can tighten by doing nothing*. [Online]
Available at: <https://www.omfif.org/2022/04/central-banks-can-tighten-by-doing-nothing/>

Williams, W., 2025. *Timeline of U.S. Stock Market Crashes*. [Online]
Available at: <https://www.investopedia.com/timeline-of-stock-market-crashes-5217820>

WRDS, W. R. D. S., 2026. *Analytics*. [Online]
Available at: <https://wrds-www.wharton.upenn.edu/pages/analytics/>
[Accessed May 2025].